OSINACHI CHUKWUJAMA

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PROFILE

Photonics master's student and software engineer with a First Class engineering degree and several years of production experience on large codebases. I implement generative models from first principles in PyTorch, from the mathematical derivation through training and sampling, and I have shipped models to edge GPUs and multimodal LLM features to production. Comfortable working independently, writing fault-tolerant Python, and explaining technical work to any audience.

RESEARCH INTERESTS

Diffusion models, flow matching, score-based models, world models, optimisation.

EDUCATION

Erasmus Mundus Joint Master, Intelligent Photonics (IPSRS) September 2026 – 2028
M1, Optics, Image, Vision and Multimedia, Université Jean Monnet, Saint-Étienne, France
Relevant coursework: Fourier optics, digital image processing, probability and signal analysis.

Bachelor of Engineering, Mechanical Engineering November 2017 – August 2023
Federal University of Technology, Owerri (FUTO), Nigeria. CGPA: 4.56/5.00 (First Class),
96th percentile in the department; best graduating student in Industrial and Production Engineering.

SKILLS

Machine learning	PyTorch: DDPM, OT-conditional flow matching, score matching, CNN/ResNet training, transfer learning (YOLOv5, ResNet18); ONNX export, TensorRT optimisation
Mathematics	Probability, variational inference (ELBO), continuity equation and optimal transport for flow matching, Fourier analysis
Software	Python (proficient), Go, TypeScript/JavaScript, PHP; Celery, RabbitMQ, PostgreSQL, GraphQL, Linux
Engineering practice	CI/CD pipelines, automated testing, typed codebases, logging and monitoring, backups and incident response
Languages	English: C1 (IELTS 8.0)

RESEARCH AND ML PROJECTS

Generative Modelling from First Principles 2026
Self-directed research

- Derived and implemented DDPM (forward/reverse process, ELBO, reparameterisation) and OT-conditional flow matching in PyTorch, covering training and sampling end to end.
- Studied the links between score matching (denoising and implicit variants, NCSN), flow matching and normalising flows; wrote a forward-diffusion noising tool in Go for synthetic data generation.
- Proposed a comparison of OT-CFM and DDPM on photonics/laser data (TPU Research Cloud application).

Margin: Handwritten Notes with Multimodal AI Feedback 2026
Personal project, margin.osinachi.me

- Built and deployed a pen-based note-taking app in which a multimodal LLM checks handwritten reasoning from images and returns immediate correctness feedback.
- Switched LLM providers to cut feedback latency; notes sync to a Git repository through a pull-then-push workflow.

Surveillance Drone with Real-Time Object Detection August 2023 – March 2024
Undergraduate final project and follow-up, supervised by Prof. O.O. Obiukwu *MEE, FUTO*

- Collected and annotated a three-class dataset, fine-tuned YOLOv5 (70% mAP) and ran live inference on a drone's camera stream via ONNX.
- Deployed the model on an NVIDIA Jetson Nano with TensorRT, cutting processing time by 44%.

Optics Simulation Playground In progress
Personal project, pre-zeemax.vercel.app

- 2D ray-tracing tool with finite-size lenses, being extended into a Fourier optics simulator.

WORK EXPERIENCE

Dstrct Group March 2025 – May 2026
Backend Engineer / Technical Lead *Lagos, Nigeria*

- Core engineer and technical lead on an enterprise intranet: a typed GraphQL backend (NestJS, TypeORM, PostgreSQL) of nearly 1,000 files spanning auth, documents, ticketing, chat, tasks, scheduling and WebSocket real-time features.
- Worked directly with clients and stakeholders to set priorities and turn business requirements into engineering tasks.
- Built an automated DOCX templating and PDF conversion pipeline generating formal documents from database records.

Hotels.ng June 2024 – January 2026
Software/DevOps Engineer (part-time) *Lagos, Nigeria*

- Developed a highly concurrent Python/Celery application distributing AI workloads across workers over RabbitMQ.
- Wrote CI/CD workflows for deployment, database migrations and automated testing.
- Wrote logging and log-processing tooling that improved monitoring coverage by over 70% versus manual checks.

Generative CAD December 2021 – June 2022
Robotics & 3D Printing Industrial Trainee *Lagos, Nigeria*

- Developed ROS path planning for robots, simulated in Gazebo and RViz; designed and 3D printed prototypes from customer requirements.

PUBLICATIONS

Quantum Computing in Artificial Intelligence: A Review of Quantum Machine Learning Algorithms, Path of Science. doi:10.22178/pos.117-25

Secure and Resilient Industrial IoT Architectures for Smart Manufacturing: A Comprehensive Review, JERR (2025). doi:10.9734/jerr/2025/v27i61548

Clinical Applications of 3D-Printed Medical Implants, JOPIR. doi:10.69613/22dn3m33

TEACHING AND COMMUNICATION

Microsoft Certified Trainer (Azure and AI): delivered enterprise workshops, including for MTN Nigeria, and prepared AI course materials.

Former Microsoft Learn Student Ambassador: trained over 300 students in software engineering and cloud technologies.

Director of Research, Mechanical Engineering, FUTO: ran hands-on training in machine learning, microcontrollers and 3D printing for 50+ students and staff. Co-founder of Hack4FUTO.

Technical writer at blog.osinachi.me.